Dhruv Venkat

Waterloo, ON | (548)-883-0423 | dvenkat@uwaterloo.ca | LinkedIn | GitHub | www.dhruvvenkat.com

EDUCATION

University of Waterloo **Sep 2024 - 2029**
Bachelor of Applied Science, Computer Engineering (Honours) *Waterloo, ON*

- **Coursework:** Data Structures and Algorithms, Digital Computers, Linear Algebra for Engineering, Electronic Circuits

TECHNICAL PROFICIENCIES

- **Languages**: TypeScript, Python, Java, Embedded C, C++, Apache Groovy, GraphQL, SQL, RISC-V Assembly, VHDL, Go
- **Libraries and Frameworks**: Node.js, ReactJS, Express.js, Next.js, Jest, JUnit 5, Scikit-Learn, TensorFlow, Pandas, Flask, FastAPI
- **Cloud Technologies**: AWS (EC2, S3, IAM, RDS, Lambda, DynamoDB), Microsoft Azure (Function Apps, Storage Blogs, AI Foundry, CosmosDB)
- **Developer Tools/Databases**: Git, Jenkins, Postman, Gradle, MySQL, SQLite, Redis, MongoDB, PostgreSQL

PROFESSIONAL EXPERIENCE

Scotiabank **Sep 2025 - Dec 2025**
Software Engineering Intern | LangChain RAG, FAISS, Tesseract OCR, Python, TypeScript, Java *Toronto, ON*

- Designed and deployed AI-powered RAG agents using LangChain and FAISS to extract business requirements from user stories, **reducing requirements-gathering time by 40% and increasing developer productivity by 25%**
- Implemented a Tesseract OCR and AI-driven parsing pipeline, **achieving a 60% reduction in manual compliance review time and 3× document throughput** by automating data extraction and signature verification of regulatory forms
- **Achieved 30% faster runtime and 50% reduction in maintenance effort for 8 enterprise applications** by upgrading backend platforms from Java 11/17 to Java 21, enabling long-term LTS support
- Enhanced application security posture, **reducing high-risk vulnerability exposure by 65% across three repositories, by monitoring 100+ vulnerabilities** and remediating issues with Checkmarx One and Black Duck

Tangerine **Jan 2025 - Apr 2025**
DevOps Engineering Intern | Jenkins, Python, Apache Groovy, RESTful APIs, CI/CD, Jira, Confluence, RHEL *Toronto, ON*

- Implemented a company-wide GitHub Copilot license tracker, **reducing monthly license spend by $1,000+** by developing, deploying, and documenting an automated solution **using RESTful APIs in Apache Groovy**
- Deployed Python-based scanners **across 100+ GitHub repositories, achieving a 20% reduction in high-severity vulnerabilities by deploying concurrent RHEL scripts** and automating detection and prioritization
- Designed and maintained **five multi-branch Jenkins CI/CD pipelines, increasing build reliability by 30%** through automated test suites, multi-branch strategies, and deployment orchestration
- Streamlined Jira and Confluence support, **reducing ticket resolution time by 20% for 10+ development teams** by triaging and troubleshooting issues, creating projects, and enforcing access policies

PROJECTS

flowstate - AI-Integrated IDE for the Terminal | Link **Apr 2026 - May 2026**
C++, POSIX TTY, CMake, Git, OpenAI/Codex

- Architected a terminal-native code editor with r**aw-mode input handling, custom rendering, redo/undo history, Git-aware change visualization, a built-in file picker**, **and AI-assisted features**
- Implemented **multi-language syntax highlighting, error detection, and project root resolution** via Language Server Protocol integrations for C/C++, Python, JavaScript/TypeScript, Go, and Rust
- Built an **integrated AI coding assistant** using Codex, enabling **editor-aware code generation**, contextual explanations, and **in-place code edits** directly in the terminal workflow

LoadStar - Deterministic API Load Simulator | Link **Jan 2026**
TypeScript, Node.js, Undici

- Architected a deterministic, phase-driven API load simulation engine supporting Poisson-distributed traffic and weighted endpoint selection, sustaining **1,200+ RPS with connection pooling and bounded concurrency while maintaining <5% client-side overhead**
- Implemented histogram-based latency aggregation (p50/p95/p99) and per-endpoint breakdown using constant-memory bucketization, enabling **real-time tail-latency analysis under 10M+ sampled requests** without memory growth
- Modeled retry amplification with exponential backoff and jitter to simulate cascading failure scenarios, demonstrating up to **2.8× effective load amplification** and identifying saturation thresholds via **automated SLO gating (p95 latency / error-rate enforcement in CI)**